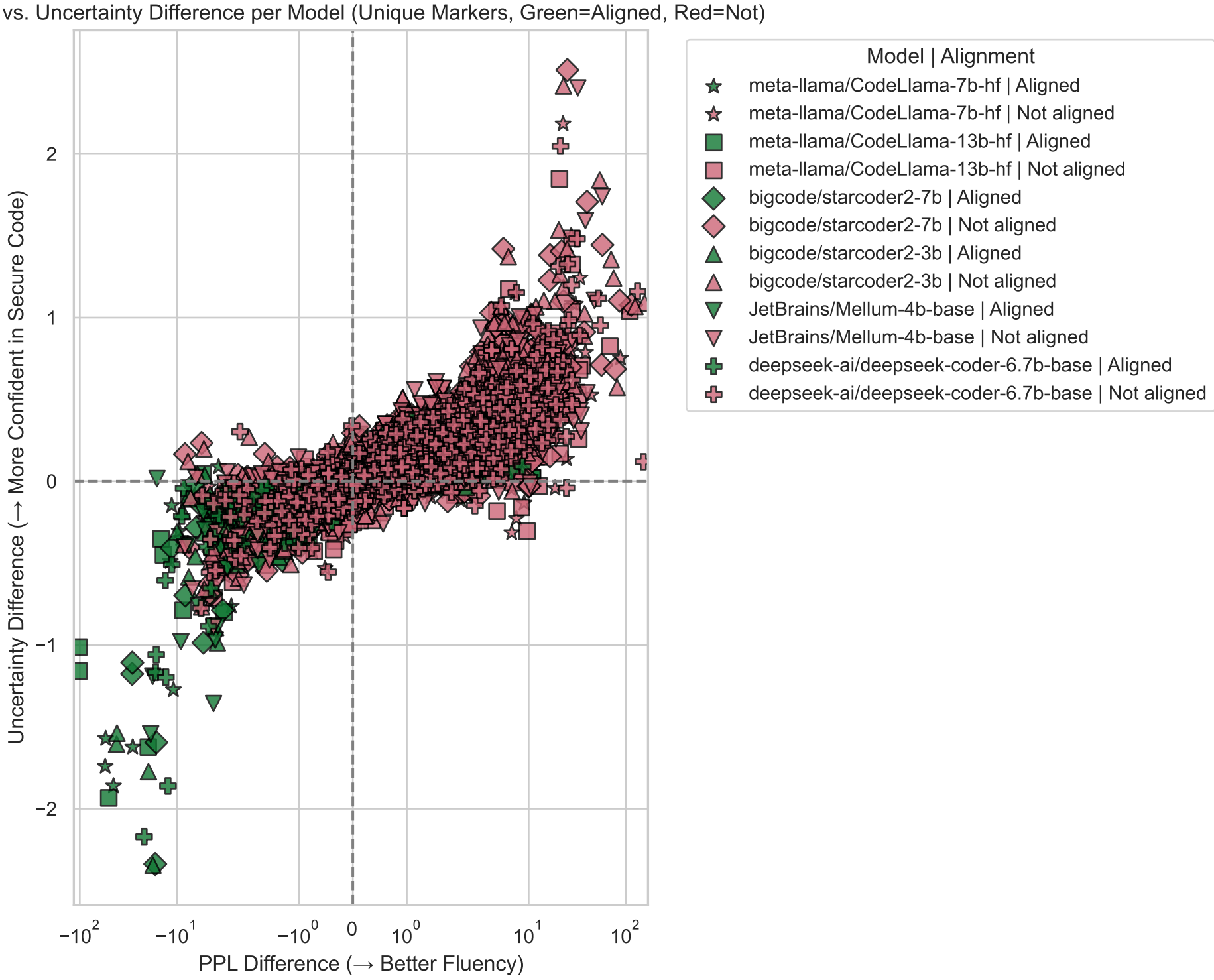
PPL vs. Uncertainty Difference per Model (Unique Markers, Green=Aligned, Red=Not)
Uncertainty Difference (→ More Confident in Secure Code)
PPL Difference (→ Better Fluency)
2
1
0
−1
−2
-10^2
-10^1
-10^0
0
10^0
10^1
10^2
Model | Alignment
meta-llama/CodeLlama-7b-hf | Aligned
meta-llama/CodeLlama-7b-hf | Not aligned
meta-llama/CodeLlama-13b-hf | Aligned
meta-llama/CodeLlama-13b-hf | Not aligned
bigcode/starcoder2-7b | Aligned
bigcode/starcoder2-7b | Not aligned
bigcode/starcoder2-3b | Aligned
bigcode/starcoder2-3b | Not aligned
JetBrains/Mellum-4b-base | Aligned
JetBrains/Mellum-4b-base | Not aligned
deepseek-ai/deepseek-coder-6.7b-base | Aligned
deepseek-ai/deepseek-coder-6.7b-base | Not aligned